

ASSIGNMENT-2

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Case study-2 : IRCTC Software

Black Box Testing for Improved Ticket Booking:

Scenario: User books a train ticket on the IRCTC platform.

Black Box Testing:

- *Objective:* Ensure the ticket booking feature functions as intended.
- 1. **Equivalence Partitioning:**
 - *Inputs:* Valid and invalid data for passenger details, journey details, and payment information.
 - *Expected Output:* Successful booking or appropriate error messages for invalid inputs.
- 2. **Boundary Value Analysis:**
 - *Inputs:* Test with minimum and maximum allowed values for ticket quantity, passenger details, and payment amount.
 - *Expected Output:* Ensure the system handles extreme values correctly.
- 3. **Decision Table Testing:**
 - *Conditions:* Payment success/failure, seat availability, valid/invalid user details.
 - *Actions:* Book ticket, show error message, redirect to payment gateway.
 - *Expected Output:* Correct action for each combination of conditions.

White Box Testing for Live Tracking of Trains:

Scenario: User tracks a train in real-time on the IRCTC platform.

White Box Testing:

- *Objective:* Ensure the live tracking feature is accurate and responsive.
- 1. **Code Coverage:**
 - *Target areas:* Live tracking algorithms, database queries for train positions.
 - *Tests:* Ensure every line of code related to live tracking is executed during testing.
- 2. **Path Testing:**
 - *Paths:* User request for train tracking -> Retrieval of live data -> Display on the user interface.
 - *Tests:* Test each possible path to ensure correct data retrieval and display.
- 3. **Performance Testing:**
 - *Load Testing:* Simulate multiple users tracking trains simultaneously to check system performance.
 - *Stress Testing:* Increase the load beyond the expected number of users to identify system limitations.

System and Security Testing for Complaint Lodging and Resolution:

Scenario: User lodges a complaint on the IRCTC platform.

System and Security Testing:

- *Objective:* Ensure the complaint lodging and resolution mechanism is secure and functional.
- 1. **Security Testing:**
 - *Authentication Testing:* Verify that only authorized users can log complaints.
 - *Data Encryption:* Ensure sensitive complaint details are stored and transmitted securely.
- 2. **Functional Testing:**
 - *User Interface Testing:* Test the complaint submission form and its fields.
 - *Workflow Testing:* Check the entire process from complaint submission to resolution.

Architecture Style and Design Properties:

Microservices Architecture:

- *Reasoning:* This allows modular development and scaling of individual features, making it easier to manage and update specific functionalities without affecting the entire system.

Scalability:

- *Reasoning:* The system should handle varying loads during peak booking times. Scalability ensures the system can accommodate increased user activity.

Security Measures:

- *Reasoning:* Given the sensitive nature of user data and financial transactions, implementing robust security measures such as encryption, secure APIs, and regular security audits is crucial.

Responsive Design:

- *Reasoning:* Ensure the software is accessible and user-friendly across various devices and screen sizes, enhancing the user experience.

Cloud-Based Infrastructure:

- *Reasoning:* Leveraging cloud services allows for flexibility, scalability, and reliability, ensuring the system can handle varying workloads and minimizing downtime.

User-Centric Design:

- *Reasoning:* Prioritize user experience through intuitive interfaces, clear navigation, and responsive design to meet the diverse needs of users.

These strategies aim to ensure the reliability, security, and user-friendliness of the IRCTC software, aligning with the defined objectives.

